Example 6.51

Find the distance between the parallel planes $x+2y-2z+1=0$ and $2x+4y-4z+5=0$.

Solution

We know that the formula for the distance between two parallel planes $ax+by+cz+d_1=0$ and $ax+by+cz+d_2=0$ is $\delta=\dfrac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$. Rewrite the second equation as $x+2y-2z+\dfrac{5}{2}=0$.

Comparing the given equations with the general equations, we get $a=1, b=2, c=-2, d_1=1, d_2=\dfrac{5}{2}$.

Substituting these values in the formula, we get the distance

$$\delta=\frac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}=\frac{\left|1-\frac{5}{2}\right|}{\sqrt{1^2+2^2+\left(-2^2\right)}}=\frac{1}{2} \text{ units.}$$

Example 6.52

Find the distance between the planes $\vec{r}\cdot\left(2\hat{i}-\hat{j}-2\hat{k}\right)=6$ and $\vec{r}\cdot\left(6\hat{i}-3\hat{j}-6\hat{k}\right)=27$

Solution

Let $\vec{u}$ be the position vector of an arbitrary point on the plane $\vec{r}\cdot(2\hat{i}-\hat{j}-2\hat{k})=6$. Then, we have

$$\vec{u}\cdot(2\hat{i}-\hat{j}-2\hat{k})=6. \quad \text{... (1)}$$

If δ is the distance between the given planes, then δ is the perpendicular distance from $\vec{u}$ to the plane

$$\vec{r}\cdot(6\hat{i}-3\hat{j}-6\hat{i})=27.$$

Therefore, $\delta=\dfrac{|\vec{u}\cdot\vec{n}-p|}{|\vec{n}|}=\left|\dfrac{\vec{u}\cdot(6\hat{i}-3\hat{j}-6\hat{k})-27}{\sqrt{6^2+(-3)^2+(-6)^2}}\right|=\left|\dfrac{3(\vec{u}\cdot(2\hat{i}-\hat{j}-2\hat{k}))-27}{9}\right|=\left|\dfrac{(3(6)-27}{9}\right|=1$ unit.

6.8.14 Equation of line of intersection of two planes

Let $\vec{r}\cdot\vec{n}=p$ and $\vec{r}\cdot\vec{m}=q$ be two non-parallel planes. We know that $\vec{n}$ and $\vec{m}$ are perpendicular to the given planes respectively. So, the line of intersection of these planes is perpendicular to both $\vec{n}$ and $\vec{m}$. Therefore, it is parallel to the vector $\vec{n}\times\vec{m}$. Let $\vec{n}\times\vec{m}=l_1\hat{i}+l_2\hat{j}+l_3\hat{k}$

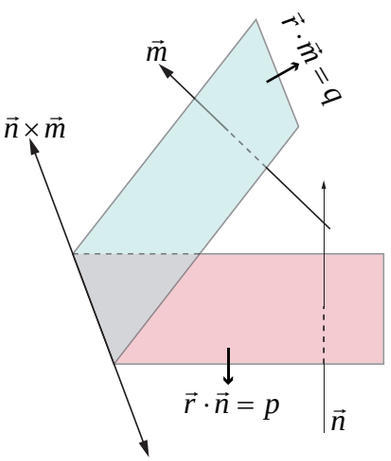


Fig. 6.33

Consider the equations of two planes $a_1x+b_1y+c_1z=p$ and $a_2x+b_2y+c_2z=q$. The line of intersection of the two given planes intersects atleast one of the coordinate planes. For simplicity, we assume that the line meets the coordinate plane $z=0$. Substitute $z=0$ and obtain the two equations $a_1x+b_1y-p=0$ and $a_2x+b_2y-q=0$. Then by solving these equations, we get the values of x and y as x_1 and y_1 respectively.

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